Somaiya Vidyavihar University · K J Somaiya School of Engineering

Cloud Computing · Mini Project Report

HeatGuard: Heatwave Forecast and Early Warning

Cloud Computing Experiments integrated with the Somaiya AI Use Case
KJS-CES-01: Climate Intelligence for Heatwave Monitoring, Prediction and Early Warning

Student	**Ayushi Ranjan**
Roll No.	16014223028
Batch	B1
Submission	Individual
Selected use case	Case 5: Climate Intelligence for Heatwave Monitoring, Prediction and Early Warning
Selected experiments	**2** Containerization, Docker and Docker Hub · **3** Application Deployment using Kubernetes · **7** Cloud Cost Optimization through Resource Right-Sizing · **8** FinOps Principles and Instance-Type Comparison
Source code	https://github.com/ayushiii2707/CC-Mini-Project---Heatwave-Forecast
Docker image	https://hub.docker.com/r/ayushiranjan27/heatguard
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Abstract

HeatGuard is a working prototype of the heatwave early-warning use case. It reads real India Meteorological Department (IMD) gridded daily maximum-temperature data for 2014–2024, forecasts the next day's maximum temperature for ten Indian cities with a machine-learning model, and converts the forecast into a heatwave alert using IMD's published criteria. On the unseen year 2024 the deployed ridge-regression model has a mean absolute error of **0.883 °C** (persistence baseline 0.94 °C), and the alert logic detects **89.6%** of the 48 actual heatwave city-days with a precision of 56.6%. The service is containerised and published on Docker Hub (Experiment 2), deployed on a Kubernetes cluster with scaling, self-healing, rollback and a rolling update (Experiment 3), right-sized from 2 vCPU / 2 GB to 0.75 vCPU / 192m with an estimated **-65.2%** change in hourly cost (Experiment 7), and compared across three instance sizes under the same workload to find the cheapest size meeting a pre-defined performance target (Experiment 8). All results in this report come from real runs; costs are estimates.

1. Problem statement, objectives and scope

Problem. Heatwaves in India are increasing in frequency, intensity and duration and affect public health, agriculture and economic activity. Severity varies by region, so timely, region-specific warnings are needed (Use-case document KJS-CES-01, §6).

Objectives.

1. Build an input → processing → output prototype on real IMD meteorological data (use case §11, steps 1–2).
2. Develop an AI model for region-wise maximum-temperature forecasting and evaluate it against a baseline (§8, objective 2).
3. Classify heatwave severity and produce an advisory (§8, objective 4; §11, steps 8–9).
4. Package, publish, deploy, measure and cost-optimise the service using Experiments 2, 3, 7 and 8.

Prototype scope.

In scope	Out of scope (see Limitations)
IMD 1°×1° gridded daily Tmax, 2014–2024, ten city grid cells	IoT Automated Weather Stations (hardware)
Next-day Tmax regression + IMD rule-based severity	Multi-day forecasts, spatial hotspot maps of all 7 IMD regions
Template advisories per severity level	LLM-generated advisories (listed as future work)
REST API + web dashboard, Docker, Kubernetes	Real cloud deployment and billing (costs are estimates)

2. Dataset

- **Source:** IMD Pune, "High resolution 1°×1° daily gridded maximum temperature" binary data, imdpune.gov.in/cmpg/Griddata/Max_1_Bin.html (Srivastava, Rajeevan and Kshirsagar, 2009). This is the repository named in the use-case document (§11, step 1).
- **Format:** one `.GRD` file per year, float32, 31 × 31 grid (7.5–37.5 °N, 67.5–97.5 °E), 99.9 = missing. File sizes were checked: 1,403,060 bytes for 365-day years and 1,406,904 bytes for leap years.
- **Acquisition:** `scripts/download_imd.sh`. The IMD server is slow and dropped connections (2017 needed three attempts), so the script retries and accepts only full-size files.
- **Pre-processing:** `scripts/prepare_data.py` parses the grid, extracts the cell nearest to each of ten cities (Delhi, Jaipur, Lucknow, Patna, Ahmedabad, Nagpur, Kolkata, Bhubaneswar, Hyderabad, Chennai) and

writes 40,180 daily rows. Grid orientation was verified (sea cells missing, Ladakh cold, Delhi ≈ 17 °C in January and ≈ 40 °C in May).

- **Split (time-ordered):** train 2014–2022 (32800 samples), validation 2023 (3650), test 2024 (3660).
- **No synthetic data** is used. The data is non-personal meteorological data, so no personal-data privacy issues arise.

3. System architecture and tools

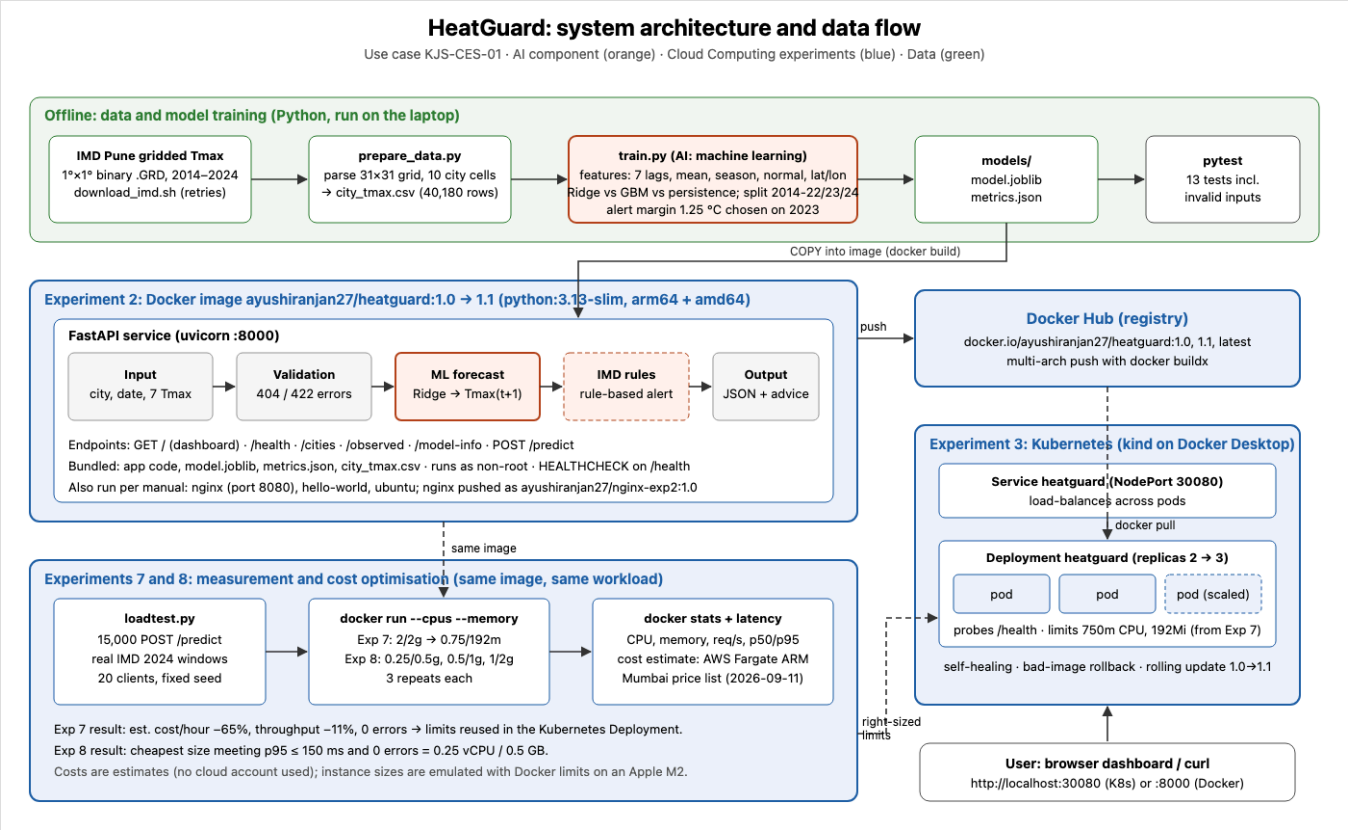


Figure 1. Labelled architecture and data flow. Orange: AI component. Blue: Cloud Computing experiments. Green: data pipeline.

Layer	Tools (versions as run)
Language / ML	Python 3.13, NumPy 2.5.4, scikit-learn 1.9.1, pandas 3.0.6 (training only), joblib 1.6.0
Service	FastAPI 0.143.0, Uvicorn 0.54.0, HTML/JS dashboard (no external libraries)
Testing / load	pytest 9.1.1, httpx 0.28.1 (async load generator)
Containers (Exp 2)	Docker Desktop, engine 29.7.2 (arm64), docker buildx, Docker Hub
Orchestration (Exp 3)	kind v0.33.0 (Kubernetes v1.37.0), kubectl v1.36.1
Measurement (Exp 7, 8)	docker stats, custom runner experiments/run_config.py, matplotlib 3.11.2
Hardware	Apple MacBook M2, 8 GB RAM (Docker VM: 8 CPUs, 3.8 GiB)

4. AI component

The AI part is kept separate from the cloud part so each can be explained on its own.

4.1 Machine-learning forecaster (next-day maximum temperature)

Input features (`app/features.py`, shared by training and serving): the last 7 daily Tmax values, their mean, day-of-year as sine/cosine, the climatological normal for the target day (from 2014–2022 only, to avoid leakage), and the cell's latitude and longitude. **Target:** Tmax on the next day. Three approaches were compared; the model with the lowest *validation* error was deployed and the test year was used only once for final reporting.

Model	Validation MAE 2023 (°C)	Test MAE 2024 (°C)	Test RMSE 2024 (°C)
persistence_baseline	0.924	0.94	1.303
ridge_regression **(deployed)**	0.888	0.883	1.225
gradient_boosting	0.895	0.902	1.248

Test MAE by city (°C): Ahmedabad 0.724, Bhubaneswar 0.869, Chennai 0.74, Delhi 1.057, Hyderabad 0.818, Jaipur 0.913, Kolkata 0.88, Lucknow 0.989, Nagpur 0.825, Patna 1.015.

Interpretation: the ML model is about 6% more accurate than the persistence baseline. Day-to-day temperature is strongly persistent, so the gain is modest but consistent; the simpler linear model generalised slightly better than gradient boosting.

4.2 Rule-based severity classifier (not ML)

`app/severity.py` applies the IMD heatwave definitions (IMD, *Heat wave definitions*): a heatwave is considered only if Tmax ≥ 40 °C (plains) or ≥ 37 °C (coastal). Departure from normal 4.5–6.4 °C → Heatwave, > 6.4 °C → Severe heatwave; for plains, Tmax ≥ 45 °C → Heatwave and ≥ 47 °C → Severe. Coastal stations have only a Heatwave category. Kolkata and Chennai are treated as coastal.

4.3 Alert margin (finding and fix)

Evaluated directly on the forecast, alerts detected only **29.2%** of real heatwave days: regression pulls extreme values towards the mean, so the forecast rarely crossed the IMD thresholds. Because a missed heatwave is costlier than a false alarm, alerts are issued on an **upper-bound forecast** (forecast + margin). The margin was chosen on the 2023 validation year by F2 score (which weights recall above precision): margin 1.25 °C. The 2023 year contains only 14 heatwave city-days, so this choice is uncertain.

Alert method (test 2024, 48 actual heatwave city-days)	Recall	Precision	F2
ML forecast + 1.25 °C margin **(deployed)**	0.896	0.566	0.802
ML forecast, no margin	0.292	1.0	0.34
Persistence baseline	0.688	0.647	0.679

Confusion matrix of the deployed alert on 2024 (rows = actual, columns = predicted; Normal / Heatwave / Severe): [3579, 33, 0], [5, 37, 2], [0, 3, 1]. The system detects most heatwave days but about 4 in 10 alerts are false alarms.

5. Implementation of the four experiments

Experiment 2: Containerization, Docker and Docker Hub

Aim (manual): install Docker, pull and run images from Docker Hub, modify containers to run an application and push it to Docker Hub. **Use-case connection:** the HeatGuard forecast service, trained model and IMD city data are packaged as one image, which every later experiment uses unchanged.

Procedure. (1) Docker Desktop installed; Docker Hub account `ayushiranjan27` created and `docker login` done by the student. (2) Manual tasks: nginx pulled and run on port 8080, hello-world and Ubuntu run, Docker commands explored. (3) Dockerfile written (python:3.13-slim, non-root user, HEALTHCHECK). (4) Image built, run and tested, including invalid inputs. (5) Image pushed for arm64 and amd64 with `docker buildx`; nginx pushed. (6) Local tag removed, image pulled back from Docker Hub and run.

Dockerfile

```
# HeatGuard: heatwave forecast API (Experiment 2: containerisation)
FROM python:3.13-slim

WORKDIR /srv
ENV PYTHONDONTWRITEBYTECODE=1 PYTHONUNBUFFERED=1

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# Application code, trained model and the processed IMD city series used by /observed
COPY app/ app/
COPY models/ models/
COPY data/processed/ data/processed/

# Run as an unprivileged user
RUN useradd --create-home appuser
USER appuser

EXPOSE 8000
HEALTHCHECK --interval=15s --timeout=3s --start-period=10s \
  CMD python -c "import urllib.request; urllib.request.urlopen('http://localhost:8000/health')"
CMD ["uvicorn", "app.main:app", "--host", "0.0.0.0", "--port", "8000"]
```

exp2_run.log

```
$ docker run -d --name heatguard -p 8000:8000 heatguard:1.0
e6e7fb4dddac7b74ede30f8adc679c1737b572ca36159d79aafc88625b7fb3e5

$ docker ps --filter name=heatguard
NAMES       IMAGE           STATUS                    PORTS
heatguard   heatguard:1.0   Up 6 seconds (healthy)   0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp

$ curl -s localhost:8000/health
{"status":"ok","model":"ridge_regression","pod":"e6e7fb4dddac"}

$ curl -s "localhost:8000/observed?city=Delhi&date=2024-05-27"
{"city":"Delhi","date":"2024-05-27","recent_tmax":[42.209049224853516,41.137046813964844,40.19950866699219,41.150436
40136719,43.54515838623047,44.9291

$ curl -s -X POST localhost:8000/predict -H "Content-Type: application/json" -d '{"city":"Delhi","date":"2024-05-2
7","recent_tmax":[<values above>]}'
{
    "city": "Delhi",
    "forecast_date": "2024-05-28",
    "forecast_tmax": 44.1,
    "upper_forecast_tmax": 45.3,
    "alert_margin_c": 1.25,
    "normal_tmax": 40.6,
    "alert": "Heatwave",
    "departure_from_normal": 4.8,
    "rule": "departure +4.8 C from normal 40.6 C",
    "advisory": "Heatwave conditions likely. Avoid sun exposure 12-3 pm, drink water often, check on elderly people,
children and outdoor workers.",
    "model": "ridge_regression",
    "inference_ms": 8.15,
    "pod": "e6e7fb4dddac",
    "note": "Educational prototype. Official warnings: India Meteorological Department."
}
```

exp2_run.log

```
# Failure scenario 1: unknown city
$ curl -s -w " HTTP %{http_code}" -X POST localhost:8000/predict -d {"city":"Atlantis",...}
{"detail":"unknown city 'Atlantis'. Supported: Delhi, Jaipur, Lucknow, Patna, Ahmedabad, Nagpur, Kolkata, Bhubaneswa
r, Hyderabad, Chennai"} HTTP 404

# Failure scenario 2: implausible sensor value (99.9 = IMD missing code)
{"detail":[{"type":"value_error","loc":["body","recent_tmax"],"msg":"Value error, temperature 99.9 C is outside the
plausible range -10..55 C","input"

# Failure scenario 3: only 3 values instead of 7
{"detail":[{"type":"too_short","loc":["body","recent_tmax"],"msg":"List should have at least 7 items after validatio
n, not 3","input":[40,41,42],"ctx"
```

exp2_manual_tasks.log

```
$ docker run -d --name web-nginx -p 8080:80 nginx:latest
b7c80f153b262ff0084cc325669d72dc3b1c2341b9638a78e55df808dc980ac6

$ curl -s localhost:8080 | grep -i "<title>"
<title>Welcome to nginx!</title>

$ docker run --rm hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (arm64v8)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

exp2_push.log

```
$ docker buildx imagetools inspect ayushiranjan27/heatguard:1.0
Name:      docker.io/ayushiranjan27/heatguard:1.0
MediaType: application/vnd.oci.image.index.v1+json
Digest:    sha256:7ac1187626dbdda4e691f0b0d935e3d238056d501cfe18d3138a02bbd495207c

Manifests:
  Name:        docker.io/ayushiranjan27/heatguard:1.0@sha256:16ea477943ddcc9311aa5da8396ca93e40c00681e683eeea9354bb4
80fc2b67c
  MediaType:   application/vnd.oci.image.manifest.v1+json
  Platform:    linux/arm64

  Name:        docker.io/ayushiranjan27/heatguard:1.0@sha256:24ecb9d5a8c2acbabe78cc009ef91d55eac6736bd1f46a5db57c305
8cc086d17
  MediaType:   application/vnd.oci.image.manifest.v1+json
  Platform:    linux/amd64

  Name:        docker.io/ayushiranjan27/heatguard:1.0@sha256:c1cc990337778bec12e3686783cd053c0e2f2e2a3c890dff5cd3f8c
a784cc192
  MediaType:   application/vnd.oci.image.manifest.v1+json
```

```
exp2_push.log

$ docker rmi ayushiranjan27/heatguard:1.0
Untagged: ayushiranjan27/heatguard:1.0

$ docker pull ayushiranjan27/heatguard:1.0
1.0: Pulling from ayushiranjan27/heatguard
Digest: sha256:7ac1187626dbdda4e691f0b0d935e3d238056d501cfe18d3138a02bbd495207c
Status: Downloaded newer image for ayushiranjan27/heatguard:1.0
docker.io/ayushiranjan27/heatguard:1.0

$ docker run -d --name heatguard-hub -p 8001:8000 ayushiranjan27/heatguard:1.0
7121f06d66c9410df245e65923452da8e97dab45d6db37147e6f701995047249

$ curl -s localhost:8001/health
{"status":"ok","model":"ridge_regression","pod":"7121f06d66c9"}
```

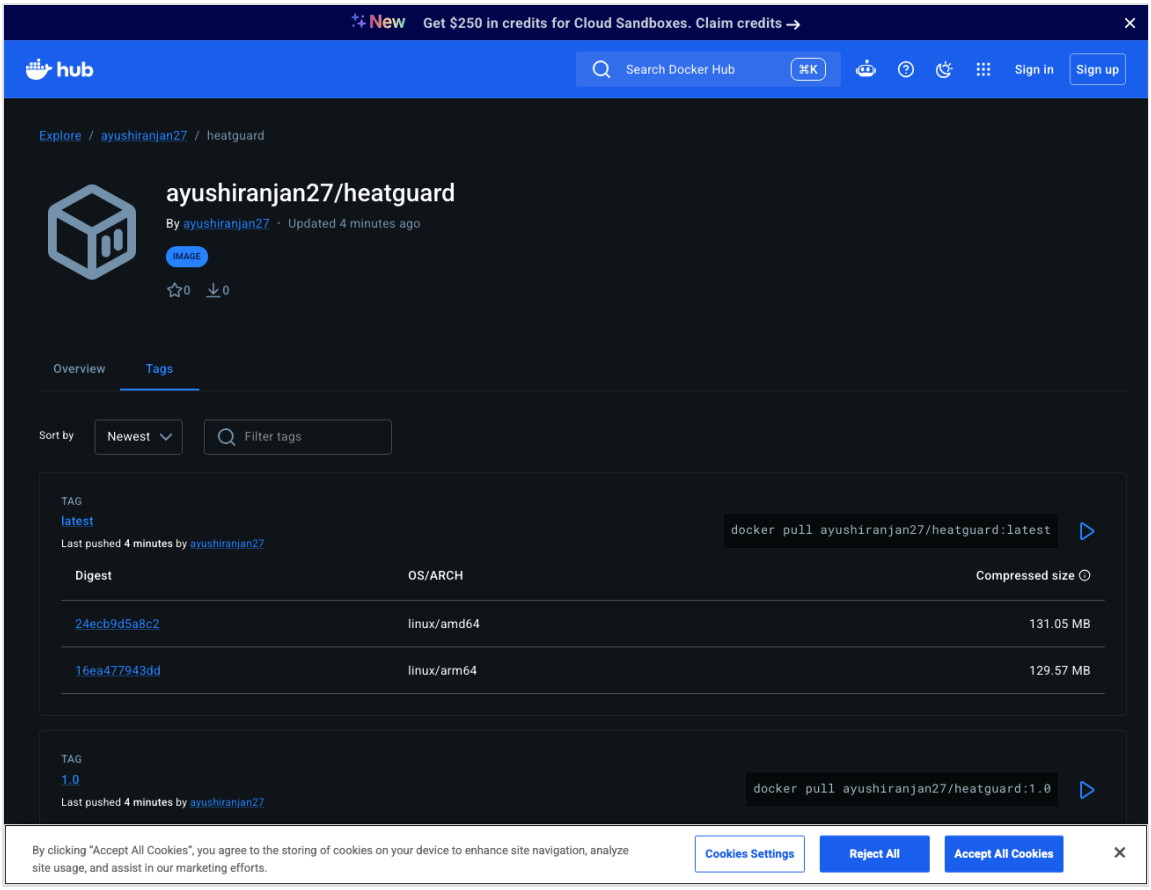


Figure 2. Docker Hub repository ayushiranjan27/heatguard with linux/amd64 and linux/arm64 images (captured after the 1.0 push).

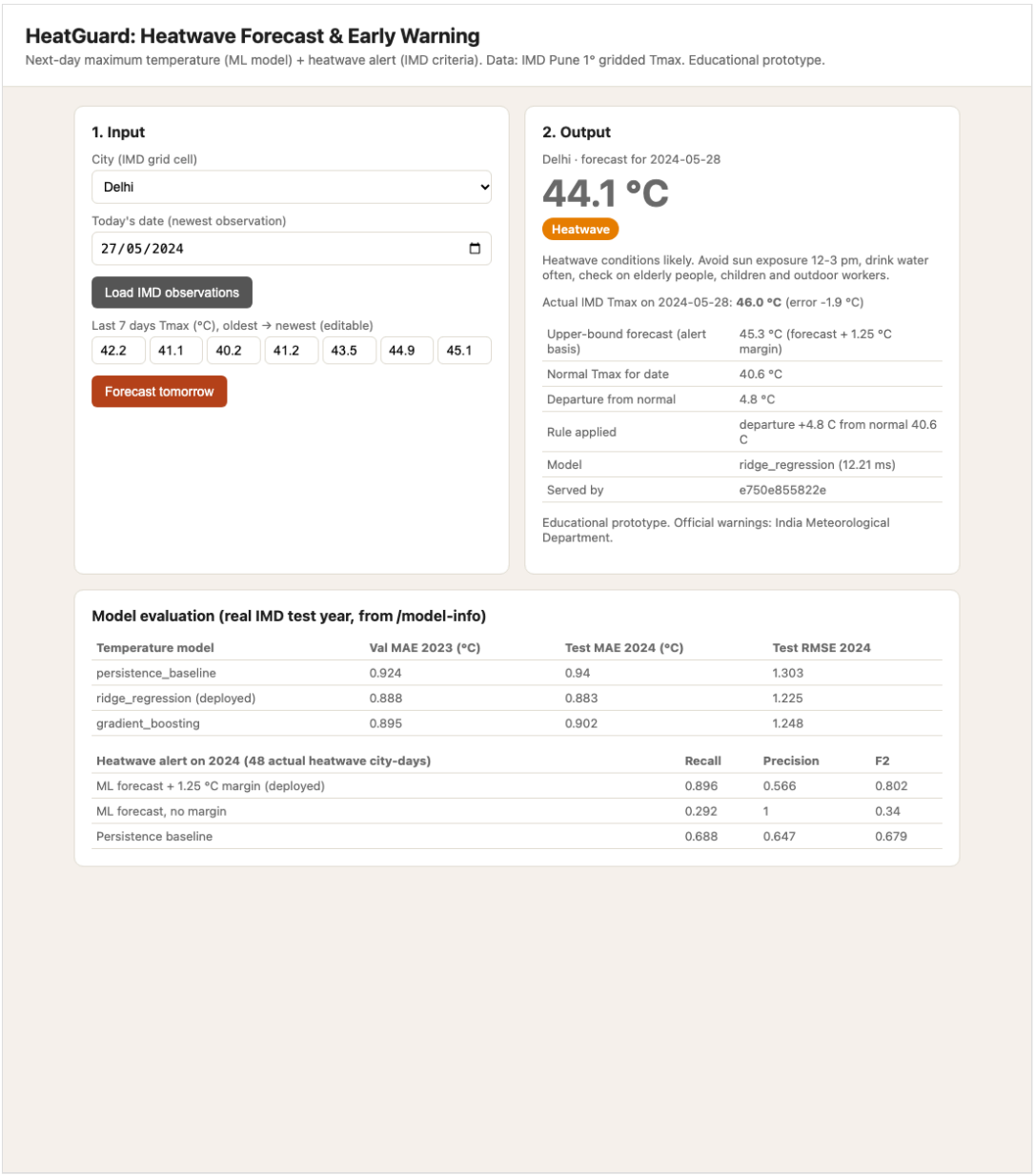


Figure 3. Dashboard served from the Docker container: Delhi, forecast for 28 May 2024 = 44.1 °C, alert Heatwave; actual IMD value 46.0 °C.

Results. Image size 136 MB (compressed content). Health check reports healthy; forecasts are served; unknown city returns 404 and malformed inputs return 422 without crashing the service. Docker Hub lists `heatguard:1.0`, `1.1` and `latest` for amd64 and arm64, and `nginx-exp2:1.0`. Version 1.1 only corrects the coastal heatwave rule (Section 4.2); forecast errors are identical.

Conclusion. The application was containerised, run and published; anyone can now run it with one `docker run` command on Intel or Apple Silicon machines.

Experiment 3: Application Deployment using Kubernetes

Aim (manual): deploy an application using Kubernetes, use kubectl commands and expose an app (preferably the image created in the Docker experiment). **Use-case connection:** the HeatGuard image from Docker Hub is deployed as a replicated, self-healing service, as an early-warning service must stay available. Resource limits come from Experiment 7.

Procedure. The manual used the Killercoda playground; this project used **kind** (Kubernetes-in-Docker), a real Kubernetes cluster running locally on Docker Desktop, so the full sequence could be scripted

(`experiments/exp3_kubernetes.sh`) and logged. Steps: create cluster → apply Deployment (2 replicas) and NodePort Service → access through the Service → scale to 3 → delete a pod → set a non-existent image tag → roll back → rolling update to 1.1.

k8s/deployment.yaml

```
# Experiment 3: HeatGuard Deployment. Image is pulled from Docker Hub (Experiment 2).
# CPU/memory limits are the right-sized values measured in Experiment 7.
apiVersion: apps/v1
kind: Deployment
metadata:
  name: heatguard
  labels: {app: heatguard}
spec:
  replicas: 2
  selector:
    matchLabels: {app: heatguard}
  template:
    metadata:
      labels: {app: heatguard}
    spec:
      containers:
        - name: heatguard
          image: docker.io/DOCKERHUB_USER/heatguard:1.1
          imagePullPolicy: Always
          ports:
            - containerPort: 8000
          resources:
            requests: {cpu: 250m, memory: 128Mi}
            limits: {cpu: 750m, memory: 192Mi}
          readinessProbe:
            httpGet: {path: /health, port: 8000}
            initialDelaySeconds: 3
            periodSeconds: 5
          livenessProbe:
            httpGet: {path: /health, port: 8000}
            initialDelaySeconds: 10
            periodSeconds: 10
```

k8s/service.yaml

```
# Experiment 3: expose HeatGuard outside the cluster on NodePort 30080.
apiVersion: v1
kind: Service
metadata:
  name: heatguard
spec:
  type: NodePort
  selector: {app: heatguard}
  ports:
    - port: 80
      targetPort: 8000
      nodePort: 30080
```

exp3_kubernetes.log

```
### 1. Deploy (image pulled from Docker Hub) and expose
$ kubectl apply -f /tmp/heatguard-deployment.yaml
deployment.apps/heatguard created

$ kubectl apply -f k8s/service.yaml
service/heatguard created

$ kubectl rollout status deployment/heatguard --timeout=300s
Waiting for deployment "heatguard" rollout to finish: 0 of 2 updated replicas are available...
Waiting for deployment "heatguard" rollout to finish: 1 of 2 updated replicas are available...
deployment "heatguard" successfully rolled out

$ kubectl get deployment heatguard
NAME        READY   UP-TO-DATE   AVAILABLE   AGE
heatguard   2/2     2            2           11s

$ kubectl get pods -o wide
NAME                         READY   STATUS    RESTARTS   AGE   IP           NODE                       NOMINATED NODE
READINESS GATES
heatguard-9ddf56c84-jxrpf   1/1     Running   0          11s   10.244.0.7   heatguard-control-plane   <none>
<none>
heatguard-9ddf56c84-wkhkq   1/1     Running   0          11s   10.244.0.8   heatguard-control-plane   <none>
<none>

$ kubectl get service heatguard
NAME        TYPE       CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
heatguard   NodePort   10.96.84.104   <none>        80:30080/TCP   10s

$ kubectl describe pod heatguard-9ddf56c84-jxrpf | grep -E 'Image:|Pulled|Limits|cpu|memory'
    Image:          docker.io/ayushiranjan27/heatguard:1.0
    Limits:
      cpu:     750m
      memory:  192Mi
    Requests:
      cpu:        250m
      memory:     128Mi
  Normal  Pulled     5s    kubelet            spec.containers{heatguard}: Successfully pulled image "docker.io/ayush
iranjan27/heatguard:1.0" in 1.465s
```

exp3_kubernetes.log

```
### 2. Service access + load balancing across pods
{"status":"ok","model":"ridge_regression","pod":"heatguard-9ddf56c84-jxrpf"}
{"status":"ok","model":"ridge_regression","pod":"heatguard-9ddf56c84-jxrpf"}
{"status":"ok","model":"ridge_regression","pod":"heatguard-9ddf56c84-wkhkq"}
{"status":"ok","model":"ridge_regression","pod":"heatguard-9ddf56c84-jxrpf"}
{"status":"ok","model":"ridge_regression","pod":"heatguard-9ddf56c84-wkhkq"}
{"status":"ok","model":"ridge_regression","pod":"heatguard-9ddf56c84-jxrpf"}

$ curl -X POST http://localhost:30080/predict (Delhi, real IMD values for 21-27 May 2024)
{"city":"Delhi","forecast_date":"2024-05-28","forecast_tmax":44.1,"upper_forecast_tmax":45.3,"alert_margin_c":1.2
5,"normal_tmax":40.6,"alert":"Heatwav

# invalid input through the Service
{"detail":"unknown city 'Atlantis'. Supported: Delhi, Jaipur, Lucknow, Patna, Ahmedabad, Nagpur, Kolkata, Bhubaneswa
r, Hyderabad, Chennai"} HTTP 404
```

exp3_kubernetes.log

```
### 4. Self-healing: delete a pod, the Deployment recreates it
$ kubectl delete pod heatguard-9ddf56c84-br798 --wait=false
pod "heatguard-9ddf56c84-br798" deleted from default namespace

$ kubectl get pods
NAME                         READY   STATUS              RESTARTS   AGE
heatguard-9ddf56c84-br798    1/1     Terminating         0          8s
heatguard-9ddf56c84-jxrpf    1/1     Running             0          19s
heatguard-9ddf56c84-ltqpr    0/1     ContainerCreating   0          0s
heatguard-9ddf56c84-wkhkq    1/1     Running             0          19s

$ kubectl get pods
NAME                         READY   STATUS    RESTARTS   AGE
heatguard-9ddf56c84-jxrpf    1/1     Running   0          35s
heatguard-9ddf56c84-ltqpr    1/1     Running   0          16s
heatguard-9ddf56c84-wkhkq    1/1     Running   0          35s

$ kubectl get deployment heatguard
NAME        READY   UP-TO-DATE   AVAILABLE   AGE
heatguard   3/3     3            3           35s

{"status":"ok","model":"ridge_regression","pod":"heatguard-9ddf56c84-ltqpr"}
{"status":"ok","model":"ridge_regression","pod":"heatguard-9ddf56c84-wkhkq"}
{"status":"ok","model":"ridge_regression","pod":"heatguard-9ddf56c84-jxrpf"}
```

exp3_kubernetes.log

```
### 5. Failure scenario: bad image tag -> rollout blocked, old pods keep serving -> rollback
$ kubectl set image deployment/heatguard heatguard=docker.io/ayushiranjan27/heatguard:does-not-exist
deployment.apps/heatguard image updated

$ kubectl get pods
NAME                         READY   STATUS             RESTARTS   AGE
heatguard-68fc79f464-vmgq5   0/1     ImagePullBackOff   0          25s
heatguard-9ddf56c84-jxrpf    1/1     Running            0          60s
heatguard-9ddf56c84-ltqpr    1/1     Running            0          41s
heatguard-9ddf56c84-wkhkq    1/1     Running            0          60s

$ curl -s http://localhost:30080/health
{"status":"ok","model":"ridge_regression","pod":"heatguard-9ddf56c84-ltqpr"}

$ kubectl rollout undo deployment/heatguard
Warning: resource deployments/heatguard was previously managed with 'kubectl apply'. Rolling back will not update th
e kubectl.kubernetes.io/last-appli
deployment.apps/heatguard rolled back

$ kubectl rollout status deployment/heatguard --timeout=180s
deployment "heatguard" successfully rolled out
```

exp3_kubernetes.log

```
### 7. Rolling update 1.0 -> 1.1 (coastal heatwave rule fix), with availability check during rollout
$ kubectl set image deployment/heatguard heatguard=docker.io/ayushiranjan27/heatguard:1.1
deployment.apps/heatguard image updated
$ kubectl rollout status deployment/heatguard
Waiting for deployment "heatguard" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "heatguard" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "heatguard" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "heatguard" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "heatguard" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "heatguard" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "heatguard" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "heatguard" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "heatguard" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "heatguard" rollout to finish: 1 old replicas are pending termination...
deployment "heatguard" successfully rolled out
# availability during rollout: 39/40 health checks returned HTTP 200 (non-200: 1)
$ kubectl get pods -o wide
NAME                         READY   STATUS    RESTARTS   AGE   IP            NODE                      NOMINATED NOD
E   READINESS GATES
heatguard-df89b7b5f-7msn6   1/1     Running   0          33s   10.244.0.12   heatguard-control-plane   <none>
<none>
heatguard-df89b7b5f-czgdj   1/1     Running   0          14s   10.244.0.14   heatguard-control-plane   <none>
<none>
heatguard-df89b7b5f-p2ck5   1/1     Running   0          21s   10.244.0.13   heatguard-control-plane   <none>
<none>
$ kubectl rollout history deployment/heatguard
deployment.apps/heatguard
REVISION  CHANGE-CAUSE
2         <none>
3         <none>
4         <none>

$ kubectl get deployment heatguard -o jsonpath="{.spec.template.spec.containers[0].image}"
docker.io/ayushiranjan27/heatguard:1.1
# coastal check through the Service (Chennai, hottest 2024 week)
[35.3, 37.2, 38.0, 39.0, 39.3, 40.8, 41.1]
```

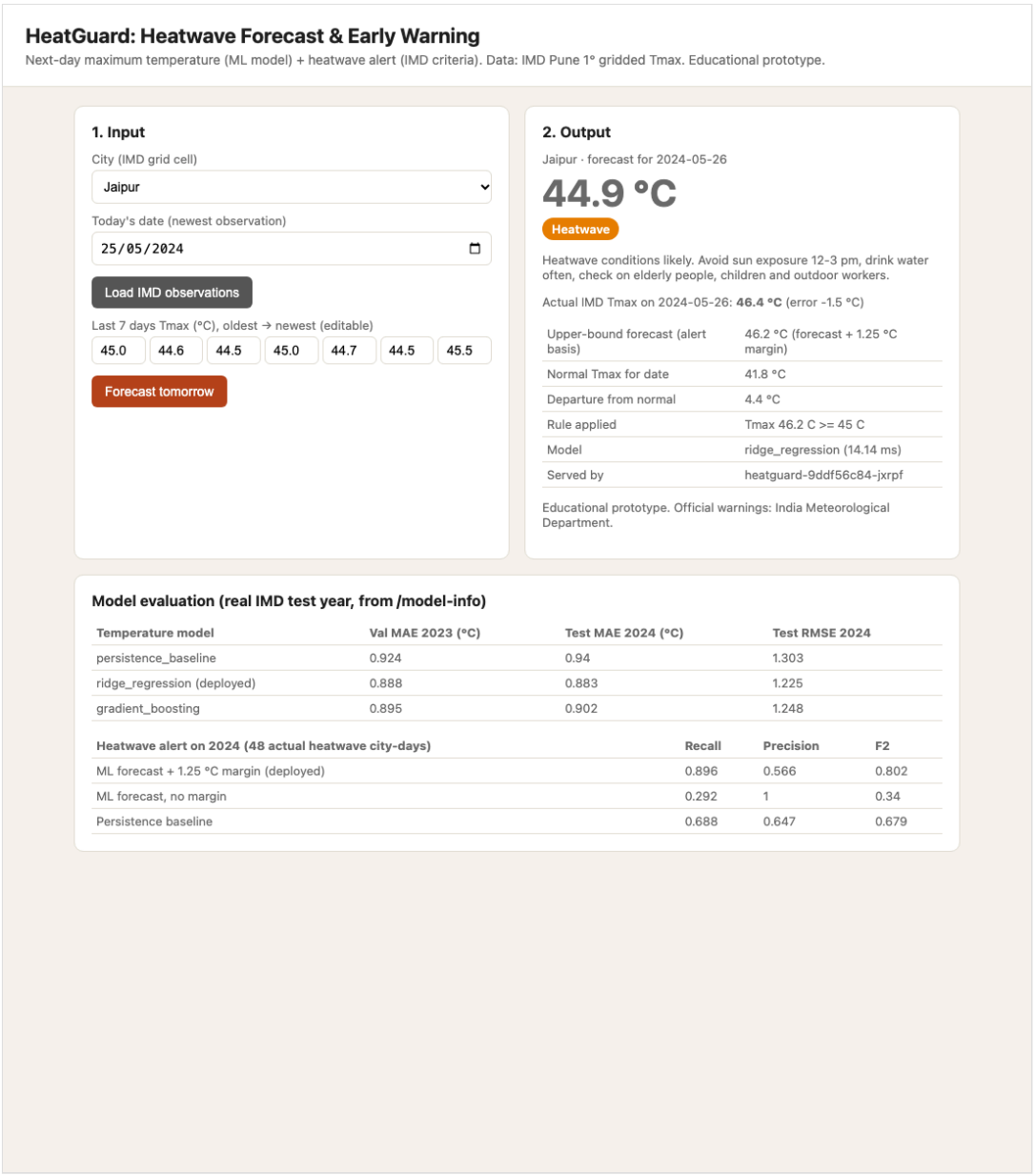


Figure 4. Dashboard through the Kubernetes NodePort Service (localhost:30080); 'Served by' shows the pod name. Jaipur, 26 May 2024: forecast 44.9 °C, Heatwave; actual 46.4 °C.

Results. The image was pulled from Docker Hub by the cluster; both pods became Ready; requests were spread across pods; scaling to 3 worked; a deleted pod was replaced by the Deployment within about 16 s; a non-existent image tag stopped in `ImagePullBackOff` while the 3 old pods kept serving, and `kubectl rollout undo` restored the Deployment. During the rolling update to 1.1, 39 of 40 health checks (every 0.5 s) returned HTTP 200 and one failed. A likely cause is a terminating pod still receiving a request before the Service endpoints were updated; a `preStop` delay is the usual remedy (not implemented).

Conclusion. Kubernetes provided declarative deployment, service discovery, scaling and recovery for the HeatGuard service.

Experiment 7: Cloud Cost Optimization through Resource Right-Sizing

Aim (manual): analyse CPU and memory utilisation of an over-provisioned workload, apply right-sizing and measure cost and efficiency improvements without performance degradation. **Use-case connection:** the workload is the HeatGuard forecast API itself, loaded with forecast requests built from real 2024 IMD data.

Procedure. (1) Run the heatguard image with 2 vCPU / 2 GB (`docker run --cpus 2 --memory 2g`). (2) Send the fixed workload: 15,000 `POST /predict` requests, 20 concurrent clients, 100 warm-up requests, fixed random seed (`loadtest/loadtest.py`), while sampling `docker stats` about once per second. (3) Repeat 3 times. (4) Right-size with the manual's rule, "slightly above peak with a safety buffer" (`experiments/rightsize.py`): CPU = peak × 1.25 rounded up to 0.25 vCPU; memory = peak × 1.5 rounded up to 64 MiB. (5) Re-run the same workload 3 times with the new limits. (6) Estimate cost.

Right-sizing decision (from `rightsizing_decision.json`): observed peak 0.566 cores and 105.5 MiB → **0.75 vCPU / 192m**.

Configuration	CPU peak (% of 1 core)	CPU used (% of limit)	Memory peak (MiB)	Memory used (% of limit)	Throughput (req/s)	p50 latency (ms)	p95 latency (ms)	Errors	Est. USD / hour	Est. USD / 1M predictions
over-provisioned 2.0 vCPU / 2g	55.1 ± 2.5	27.5 ± 1.2	105.1 ± 0.5	5.1 ± 0.1	661.3 ± 19.9	15.0 ± 0.3	97.9 ± 3.0	0	0.0529	0.0222
right-sized 0.75 vCPU / 192m	47.0 ± 1.7	62.6 ± 2.3	115.5 ± 0.8	60.1 ± 0.4	590.5 ± 12.4	16.5 ± 0.4	108.6 ± 2.0	0	0.0184	0.0086

Change after right-sizing: throughput -10.7%, p50 latency +10.0%, p95 latency +10.9%, estimated hourly cost -65.2%, estimated cost per million predictions -61.3%. No errors and no out-of-memory kills in any run. The right-sized limits are used in the Kubernetes Deployment (Experiment 3).

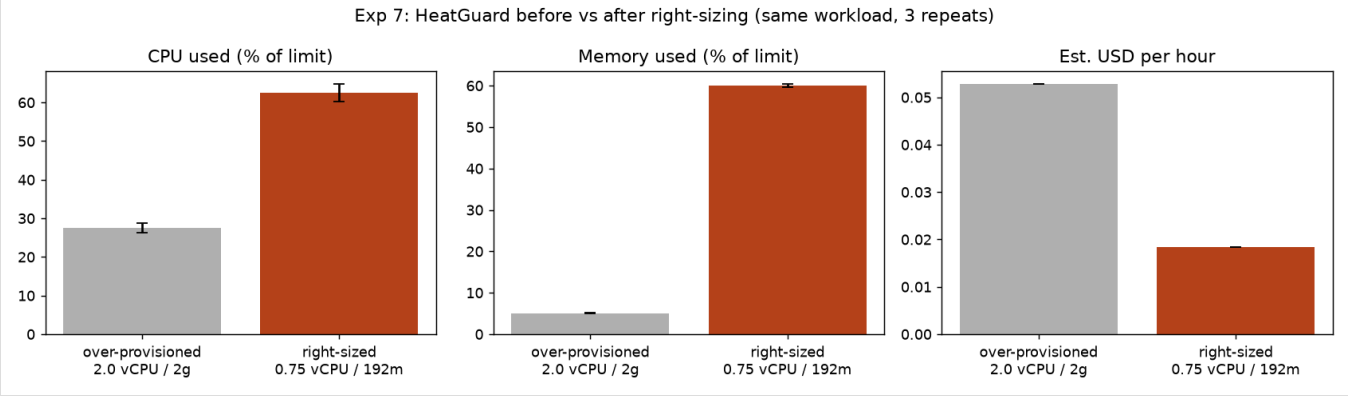


Figure 5. Experiment 7: utilisation of the allocated resources and estimated hourly cost before and after right-sizing (mean ± std, 3 runs).

Cost basis (estimate, not an actual charge): AWS Fargate on-demand, Linux/ARM, Asia Pacific (Mumbai): USD 0.02383 per vCPU-hour and USD 0.00261 per GB-hour, read from the AWS public price list (publication date 2026-09-11), saved as `experiments/pricing/aws_ecs_ap-south-1_index.json`. Hourly cost = vCPU × 0.02383 + GB × 0.00261; cost per million predictions = hourly cost ÷ (throughput × 3600) × 10^6. No cloud account was used.

Conclusion. The service was heavily over-provisioned (≈ 28% of CPU and 5% of memory used). Right-sizing cut the estimated cost by about two-thirds for a ~11% throughput reduction, an acceptable trade-off for this service. The manual's "without performance degradation" is only approximately met: a small, measured degradation occurred.

Experiment 8: FinOps Principles and Instance-Type Comparison

Aim (manual): compare different instance (resource) configurations for the same workload and identify the most cost-efficient option without performance degradation. **Use-case connection:** chooses the cheapest size on which the heatwave API still meets its response-time target.

Procedure (manual algorithm steps 1–10). Workload: identical to Experiment 7. Candidates: three AWS Fargate-valid sizes, emulated with Docker limits. The **performance target was written down before running** (`experiments/exp8_configs.json`): p95 latency ≤ 150 ms, zero errors, no OOM kill. Each configuration ran 3 times.

Configuration	CPU peak (% of 1 core)	CPU used (% of limit)	Memory peak (MiB)	Memory used (% of limit)	Throughput (req/s)	p50 latency (ms)	p95 latency (ms)	Errors	Est. USD / hour	Est. USD / 1M predictions	Meets target
low 0.25 vCPU / 512m	25.9 ± 0.5	103.4 ± 1.8	118.2 ± 0.7	23.1 ± 0.1	316.2 ± 5.1	54.8 ± 3.2	112.0 ± 0.7	0	0.0073	0.0064	yes
medium 0.5 vCPU / 1g	42.5 ± 1.9	85.0 ± 3.8	106.0 ± 0.5	10.3 ± 0.1	580.1 ± 35.9	16.7 ± 1.1	108.9 ± 6.7	0	0.0145	0.0070	yes
high 1.0 vCPU / 2g	52.9 ± 1.1	52.9 ± 1.1	105.7 ± 0.4	5.2 ± 0.1	603.2 ± 8.1	16.0 ± 0.1	107.3 ± 2.5	0	0.0290	0.0134	yes

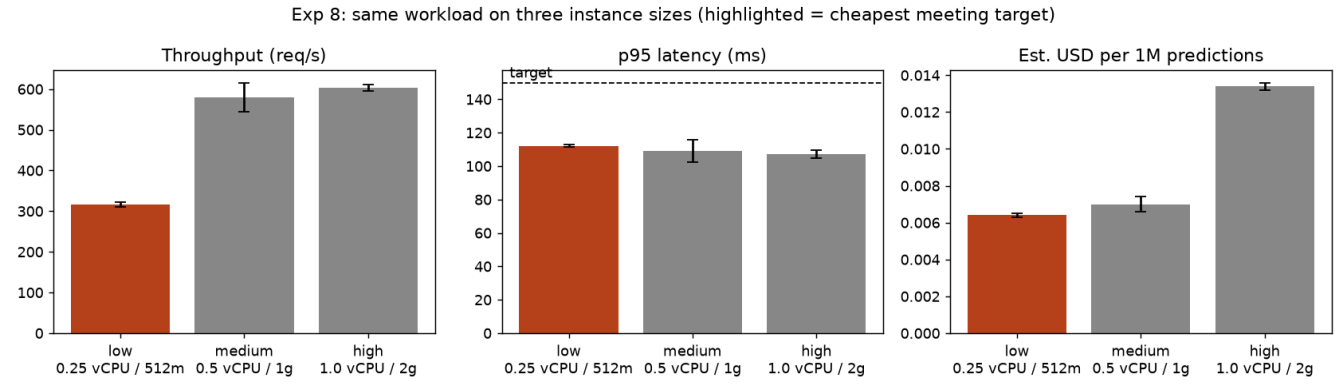


Figure 6. Experiment 8: throughput, p95 latency (dashed line = target) and estimated cost per million predictions (mean ± std, 3 runs).

Result. All three sizes met the pre-defined target; the most cost-efficient is **low (0.25 vCPU / 512m)** at an estimated USD 0.0064 per million predictions, about half the cost of the 1 vCPU size. It does so with roughly half the throughput and about 3× higher median latency. The target did not include a throughput requirement. As a *post-hoc sensitivity check* (not the pre-defined target): if ≥ 500 req/s were also required, the medium size would be chosen.

Conclusion. Doubling resources beyond 0.5 vCPU gave almost no extra performance (a single-process Python server uses at most about one core) but doubled the cost. FinOps decisions depend on stating the performance target explicitly.

6. Testing and results

Automated tests (`pytest -v`, result: **13 passed in 1.21s**):

Test	Type
test_health_reports_loaded_model	representative / rules
test_cities_list	representative / rules
test_observed_returns_real_imd_window	representative / rules
test_predict_hot_spell_raises_alert	representative / rules
test_predict_winter_is_normal	representative / rules
test_forecast_close_to_actual_on_real_day	representative / rules
test_unknown_city_404	invalid input / failure
test_wrong_number_of_values_422	invalid input / failure
test_implausible_temperature_422	invalid input / failure
test_non_numeric_and_bad_date_422	invalid input / failure
test_observed_outside_data_range_404	invalid input / failure
test_severity_rules	representative / rules
test_day_of_year_aligns_leap_and_non_leap	representative / rules

Failure scenarios demonstrated on the running container and through Kubernetes: unknown city (404), 3 values instead of 7 (422), an IMD missing-value code 99.9 °C (422), invalid date (422), a non-existent image tag (ImagePullBackOff, rolled back), and a deleted pod (recreated). **Measurements:** forecast MAE/RMSE, alert recall/precision/F2, throughput, p50/p95 latency, CPU and memory utilisation and estimated cost, summarised in Sections 4 and 5.

7. Requirement-to-test checklist

Requirement (assignment brief)	Implementation	Evidence	Status
Working prototype: input → processing → output	Dashboard + POST /predict (city, date, 7 Tmax → forecast, alert, advice)	exp2_run.log, screenshots 01–03, 05	Verified
AI/ML component appropriate to use case	Ridge regression next-day Tmax on IMD data; compared with GBM and persistence	metrics.json, train_output.txt	Verified
Rule-based part identified	IMD heatwave criteria (app/severity.py), labelled rule-based	test_severity_rules	Verified
Real / public dataset documented	IMD Pune 1° gridded Tmax 2014–2024, download script	data/processed/city_tmax.csv	Verified
Labelled architecture diagram	docs/architecture.html → 00_architecture_diagram.png	Figure 1	Verified
Exp 2: containerise, run, push to Docker Hub	Dockerfile, multi-arch push, pull-and-run check, nginx push	exp2_build/run/push logs, Docker Hub API + screenshot 04	Verified
Exp 3: deploy, expose, pod status, scaling/recovery	kind cluster, Deployment + NodePort, scale 2→3, pod delete, bad image rollback, rolling update	exp3_kubernetes.log, screenshots 03, 05	Verified
Exp 7: measure CPU/memory, right-size, compare	docker stats under fixed load, rule-based right-sizing, 3 repeats	exp7/*.csv, runs.jsonl, rightsizing_decision.json	Verified
Exp 8: same workload on configs, cheapest meeting target	3 Fargate-valid sizes, target fixed beforehand	exp8/*, exp8_configs.json, summary.md	Verified
Exp 7 & 8 same workload, cost basis stated	Same loadtest.py, seed, 15000 requests, 20 clients; AWS price list saved	pricing/aws_ecs_ap-south-1_index.json	Verified
Representative inputs tested	pytest + curl on real IMD windows (Delhi, Jaipur, Nagpur, Chennai)	pytest_output.txt	Verified
≥1 invalid-input / failure scenario	unknown city 404, 3 values 422, 99.9 °C 422, bad date 422, bad image tag in K8s	pytest_output.txt, exp2_run.log, exp3_kubernetes.log	Verified
Measurements (error, latency, throughput, CPU/mem, cost)	MAE/RMSE, recall/precision, req/s, p50/p95, docker stats, est. USD	metrics.json, summary.md	Verified
Clean-environment setup	Fresh git clone + new venv: pip install, 13/13 tests pass, retraining reproduces MAE 0.883 and margin 1.25; published image pulled and run after deleting local tag. Docker/K8s steps not repeated from scratch	exp2_push.log (pull + health); clean-clone check	Partially Verified
Demonstration video (5 min)	Plan in docs/DEMO_PLAN.md	to be recorded by student	Not Verified

8. Limitations, data privacy and improvements

- **Spatial resolution:** 1° grid cells (~100 km) are averages, not station readings; city values are approximations.
- **Forecast horizon:** one day ahead only; uses only past temperature (no humidity, wind or numerical weather model input).

- **Heatwave rules:** applied per cell per day; IMD declares heatwaves when criteria hold at ≥ 2 stations of a sub-division for 2 consecutive days. Normals come from 2014–2022 rather than IMD's 30-year station normals.
- **Alert margin:** tuned on a validation year with only 14 heatwave city-days; precision is 0.57 (many false alarms).
- **Experiments 7 and 8:** instance sizes were emulated with Docker limits on one laptop, not cloud VMs; the single Python load generator limited throughput to ~600 req/s; costs are estimates from public prices.
- **Experiment 3:** single-node local cluster; one failed health check during the rolling update.
- **Data privacy:** only public, non-personal meteorological data is processed; no user data is stored; the container runs as a non-root user; Docker Hub credentials were kept out of the code and repository (`.env.example` holds only the username variable).
- **Improvements:** add humidity/heat-index and IMD forecast inputs; multi-day forecasts; consecutive-day rule; quantile regression for calibrated uncertainty; LLM-generated multilingual advisories with human review (use case §13); Horizontal Pod Autoscaler and `preStop` hooks; repeat Experiments 7–8 on real cloud instances.

9. Conclusion

A working heatwave early-warning prototype was built on real IMD data. Its ML forecaster beats a persistence baseline, and the validated alert margin raises heatwave detection on the unseen 2024 year from 29% to 90%. The same service was containerised and published (Exp 2), deployed and operated on Kubernetes (Exp 3), right-sized for an estimated 65% cost reduction (Exp 7), and compared across instance sizes to choose the cheapest one meeting a stated target (Exp 8). Each experiment acts on the same artifact, so together they form one pipeline from model to cost-optimised deployment.

10. References

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11. Acknowledgement of AI-assisted work

This project was developed with the help of Claude (Anthropic), an AI assistant, used for analysing the assignment documents, writing code and scripts, and drafting this report. All code was executed on the student's machine; all results, logs and screenshots were produced by real runs and were checked by the student. The student created the Docker Hub account and performed the `docker login` herself. Command outputs in this report are copied verbatim from the log files in `experiments/results/`; screenshots of the dashboard and Docker Hub were captured with headless Google Chrome from the running services.